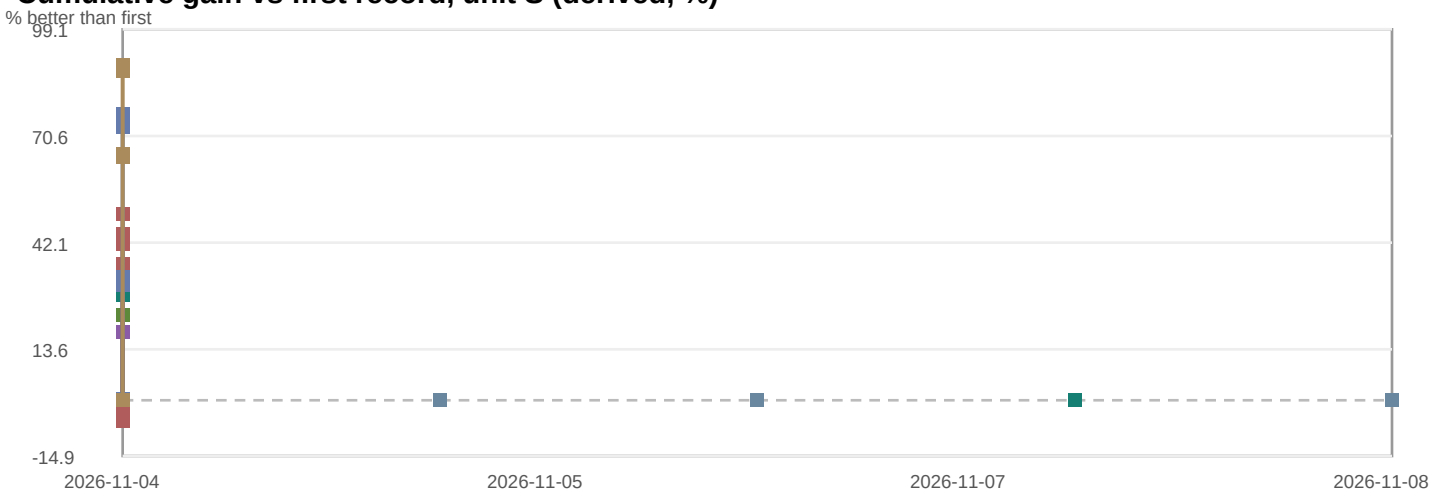
Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 50, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

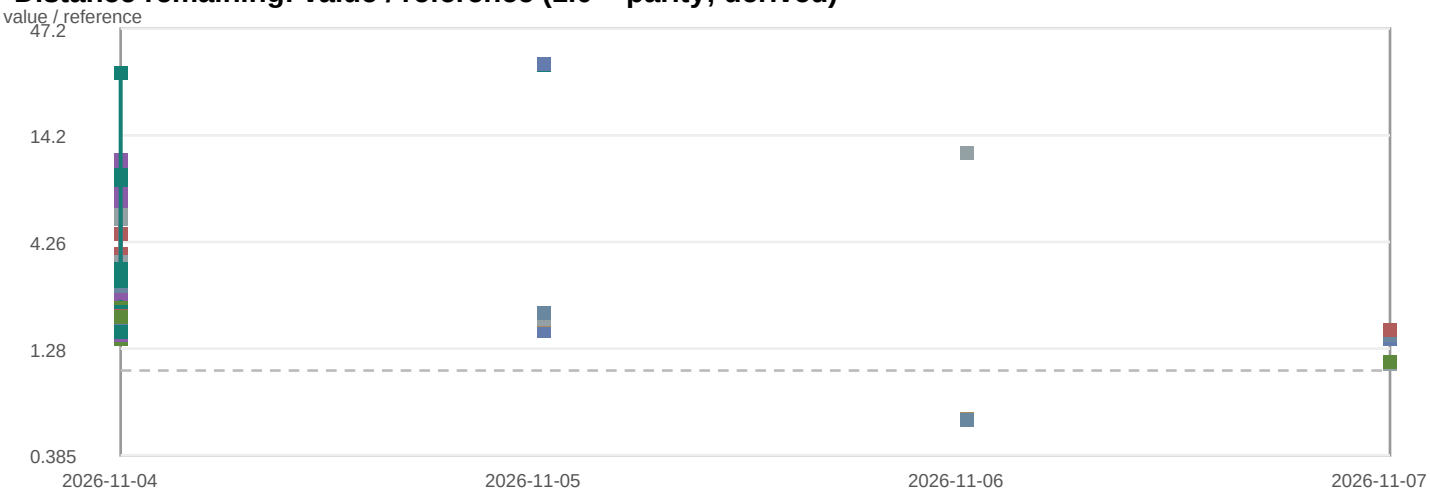
Cumulative gain vs first record, unit S (derived, %)



Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •

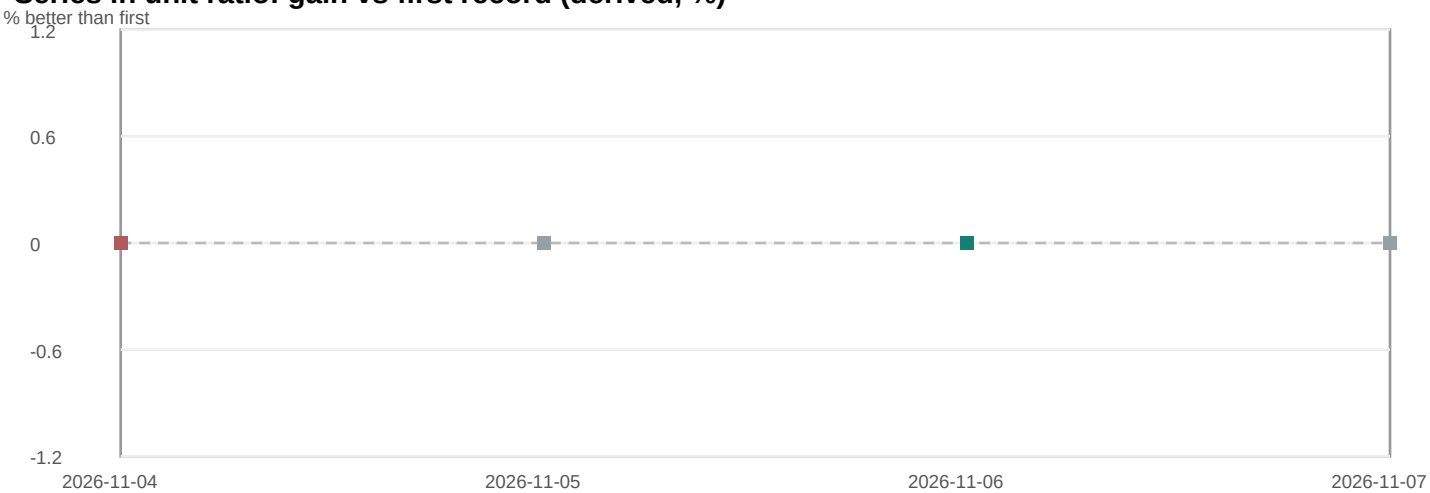
Distance remaining: value / reference (1.0 = parity; derived)



Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

Series in unit ratio: gain vs first record (derived, %)



Series key

- analysis:ecbench_statistics_v2
- analysis:ecbench_statistics_v2
- archive:ic-packed-filter-p32-2
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe

Series in unit ratio: gain vs first record (derived, %) (series key continued)

- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- kernel:f6_ic_gate • f4_control
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_word_
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p

Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
analysis:ecbench_statistics_v2 • censored_ratio_	ratio	648.2	648.2	-	-	1
analysis:ecbench_statistics_v2 • heavy_tail_rows	count	71	71	-	-	1
analysis:ecbench_statistics_v2 • ops_interval_wi	ratio	13.1	13.1	-	-	1
archive:ic-packed-filter-p32-20261029 • s_prime_	S	31.6936	31.6936	0%	31.4718	1
archive:ic-packed-filter-p32-20261029 • s_prime_	S	63.4961	63.4961	0%	31.6255	1
archive:ic-packed-filter-p32-20261029 • wall_per	ratio	0.9601	0.9601	0%	-	1
archive:icb-large-prime-n53-s1-20261029 • report	count	240	240	-	-	1
archive:icb-large-prime-n53-s2-20261029 • report	count	240	240	-	-	1
archive:icb-large-prime-n61-s1-20261029 • report	count	72	72	-	-	1
archive:icb-large-prime-n61-s2-20261029 • report	count	60	60	-	-	1
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	0.5739	0.5739	0%	0.5814	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	12.2426	12.2426	0%	11.6066	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_	S	0.5819	0.5819	0%	0.5746	2
ic:curve25519-family-k20-32 • ratio_w0_k24	ratio	0.9375	0.9375	0%	-	1
ic:curve25519-family-k20-32 • ratio_w0_k32	ratio	0.8802	0.8802	0%	-	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k20	S	2.5746	2.5746	0%	1.7601	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k24	S	2.1638	1.6735	22.6592%	1.429	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k28	S	1.9211	1.5717	18.1849%	1.4943	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k32	S	2.11	1.5152	28.1881%	1.5539	2
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.601	2.601	0%	1.5669	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.111	2.111	0%	1.7695	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.869	1.869	0%	1.7971	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.93	1.93	0%	1.9052	1
ic:curve25519-family-k20 • s_w0_incumbent	S	8.3478	4.8598	41.7827%	3.2356	5
ic:curve25519-family-k24-28 • s_w0_incumbent_k24	S	2.5842	2.5842	0%	2.0304	1
ic:curve25519-family-k24-28 • s_w0_incumbent_k28	S	2.3732	2.3732	0%	2.3865	1
ic:curve25519-family-k24-32 • best_train_s_w0_k2	S	2.713	2.713	0%	-	1
ic:curve25519-family-k24-32 • best_train_s_w0_k3	S	2.768	2.768	0%	-	1
ic:curve25519-family-k24-32 • ratio_w0_k32	ratio	0.9106	0.9106	0%	-	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k24	S	2.1055	2.1055	0%	1.942	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k28	S	1.8835	1.8835	0%	1.8154	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k32	S	1.8423	1.8423	0%	1.8232	1
ic:curve25519-family-k24 • s_w0_incumbent	S	6.607	3.3182	49.7782%	2.7163	5
ic:curve25519-family-k28-32 • s_w0_incumbent	S	2.5798	2.5798	0%	2.5891	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k28	S	1.969	1.969	0%	1.8593	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k32	S	1.781	1.781	0%	1.8229	1
ic:curve25519-family-k28 • s_w0_incumbent	S	10.9395	2.576	76.452%	2.7572	7
ic:curve25519-family-k32 • s_w0_incumbent	S	24.1377	2.519	89.5641%	2.7453	5
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2283	1.2283	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2385	1.2385	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.1542	1.1542	-0%	-	1
ic:curve25519-k20-single-target • s_f1_x07_w01	S	2.3781	2.3781	0%	1.4287	1
ic:curve25519-k20-single-target • s_f1_x07_w0	S	2.3211	2.3211	0%	1.4951	1
ic:curve25519-k20-single-target • s_f1_x07_w1	S	2.8907	2.8907	0%	1.0818	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1475	1.1475	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1636	1.1636	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.0384	1.0384	-0%	-	1
ic:curve25519-k24-single-target • s_f1_x07_w01	S	1.8945	1.8945	0%	1.4975	1
ic:curve25519-k24-single-target • s_f1_x07_w0	S	1.8341	1.8341	0%	1.5802	1
ic:curve25519-k24-single-target • s_f1_x07_w1	S	2.4383	2.4383	0%	1.1058	1
kernel:f6_ic_gate • f4_control_word_xors_off_ove	ratio	1	1	-	-	1
kernel:f6_ic_gate • f6ic_cpu_off_over_on_max	ratio	10.29	10.29	-0%	-	1
kernel:f6_ic_gate • f6ic_cpu_off_over_on_min	ratio	2.67	2.67	-0%	-	1
kernel:f6_ic_gate • f6ic_word_xors_off_over_on_m	ratio	1.84	1.84	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	0.936	0.936	-0%	-	1

kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	2.883	2.883	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	1.589	1.589	-0%	-	1
kernel:inherited_f4_solve_tree • work_par_over_s	ratio	2.865	2.865	0%	-	1
registry-rebinding:ic-aimed-gather • raw_pins_re	count	211	211	-	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	0.718	0.718	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0948	1.0948	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.4723	1.4723	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0653	1.0653	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0008	1.0008	-0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	1.0827	1.0827	0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	0.9842	0.9842	0%	-	1
rho:prime_medium_tier_bound • prime-medium/bsgs-	S	1.9684	1.9684	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.0422	1.0422	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	0.9473	0.9473	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.8961	1.8961	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.0658	1.0658	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	0.969	0.969	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-n	S	1.9367	1.9367	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1
rho:prime_medium_tier_bound • prime-medium/rho-p	S	1.2703	1.2703	0%	-	1

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.